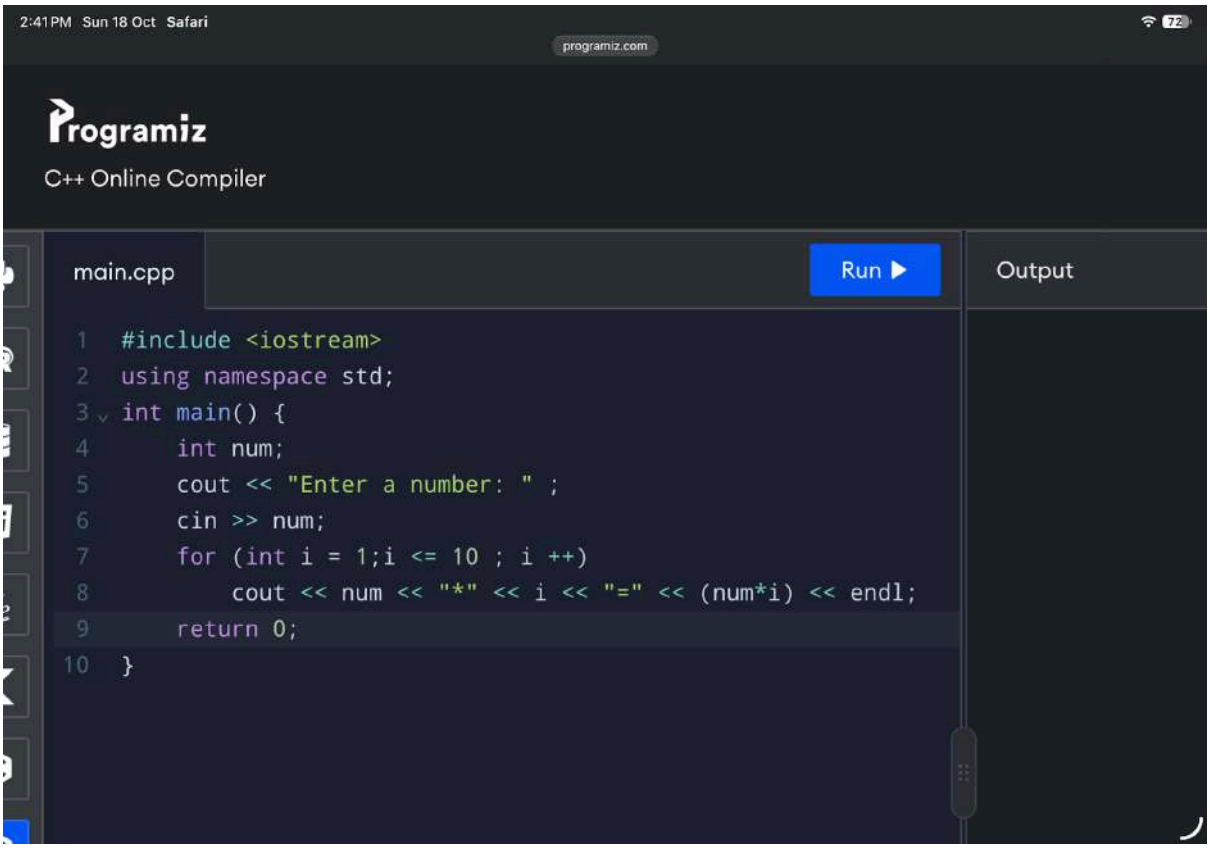
Name: Eesha Zahid
Section: BSCS- C
Roll no: 36

Task 1:

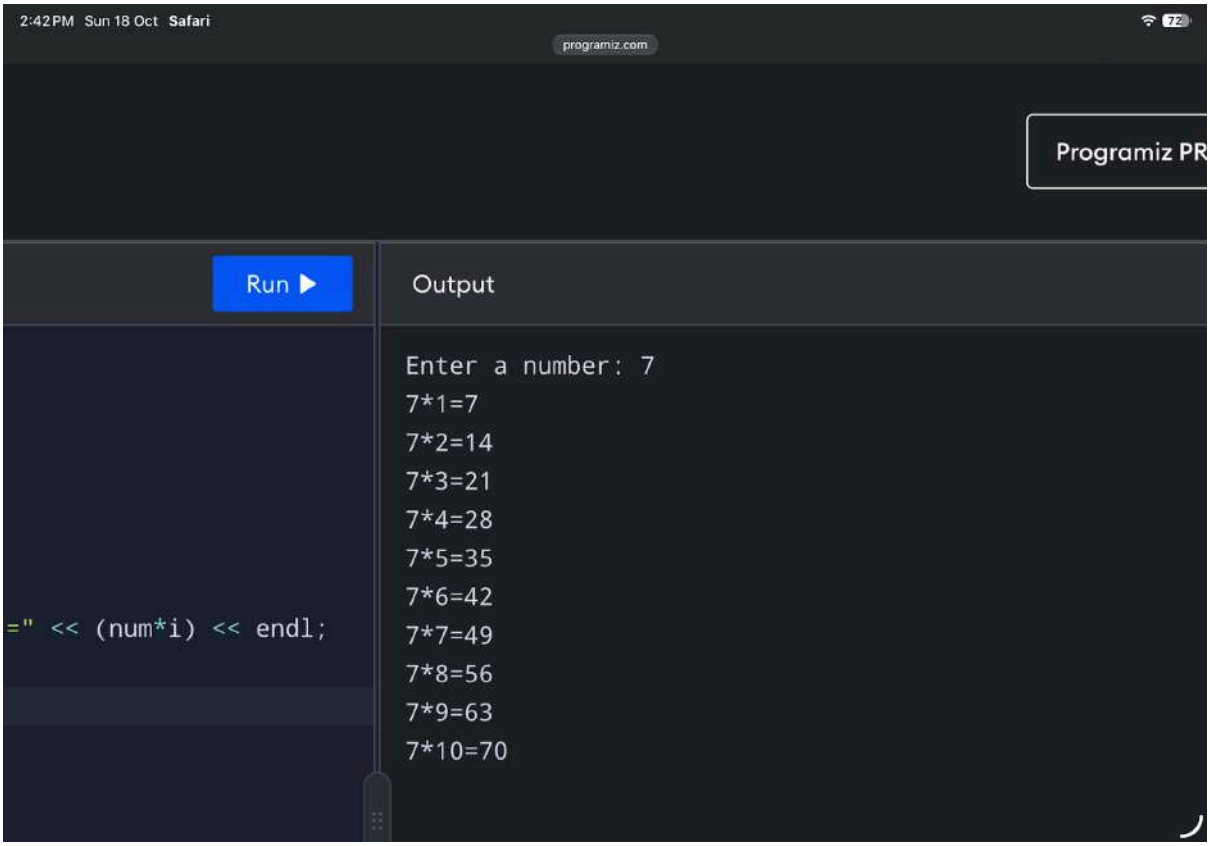
Lab 5

Task - 1

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a num:";
    cin >> num;
    for (int i = 1 ; i <= 10 ; i++)
        cout << num << "*" << i << = (num*i) << endl;
    return 0;
}
```



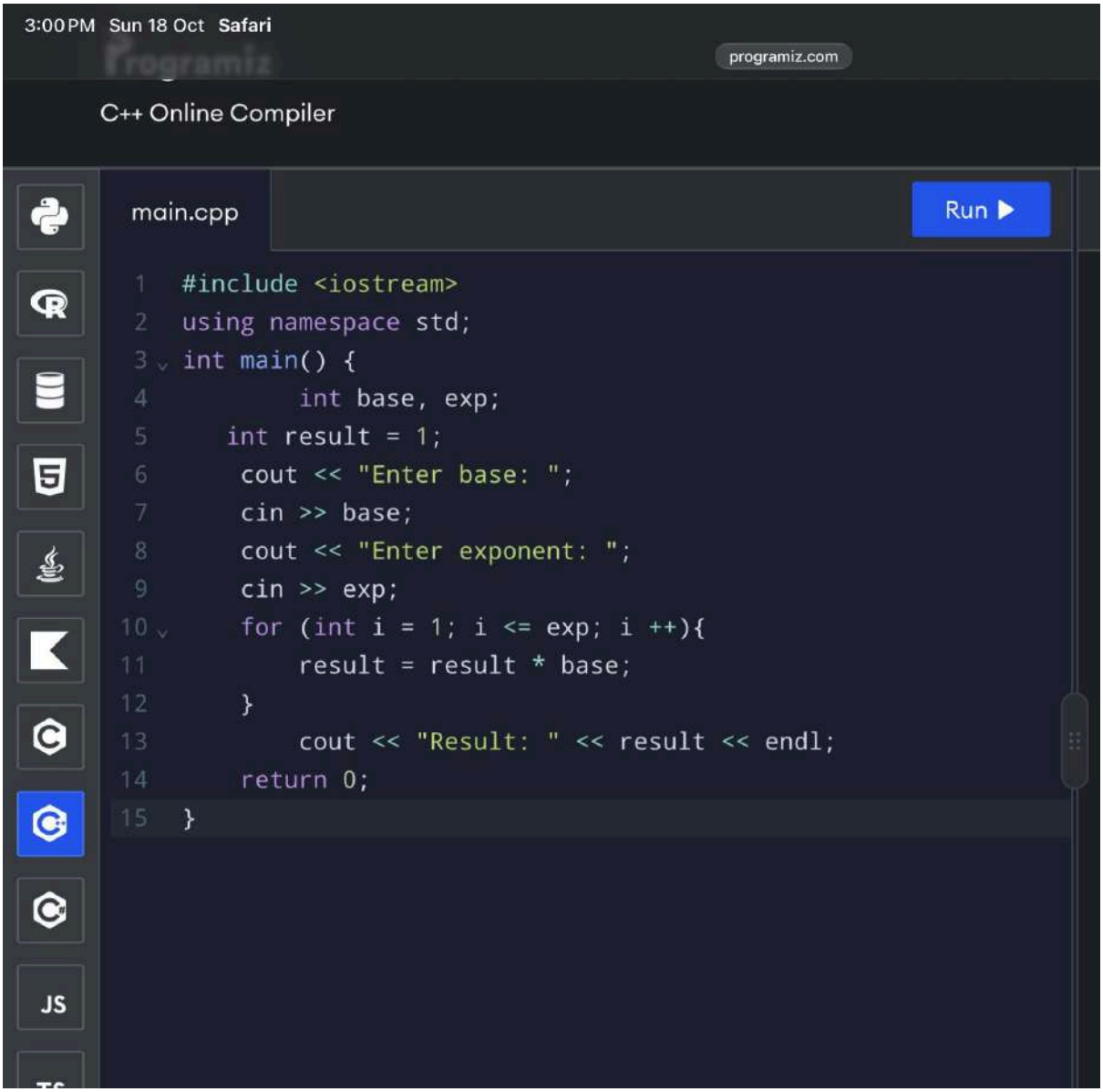
Output:


2:42 PM Sun 18 Oct Safari
programiz.com
Programiz PR
Run ▶
Output
=" << (num*i) << endl;
Enter a number: 7
7*1=7
7*2=14
7*3=21
7*4=28
7*5=35
7*6=42
7*7=49
7*8=56
7*9=63
7*10=70

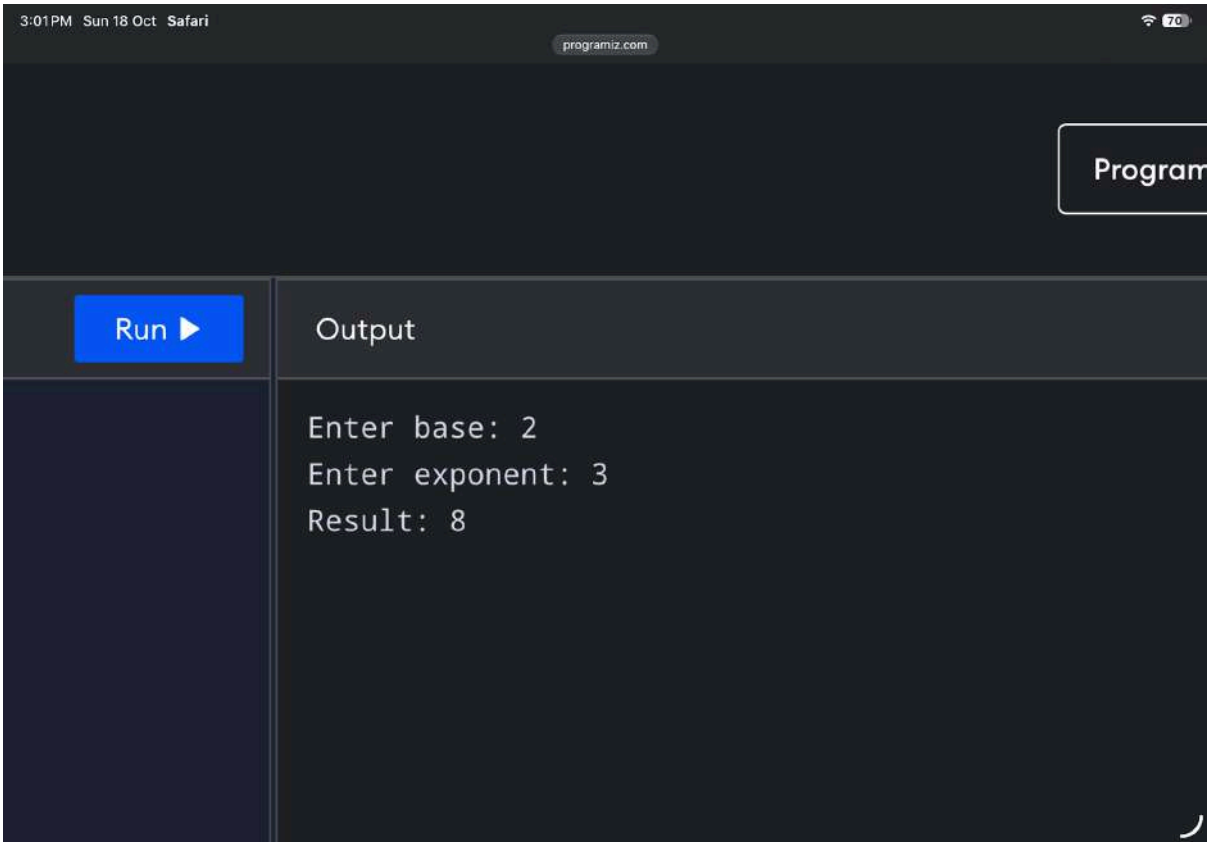
Task 2:

Task - 2

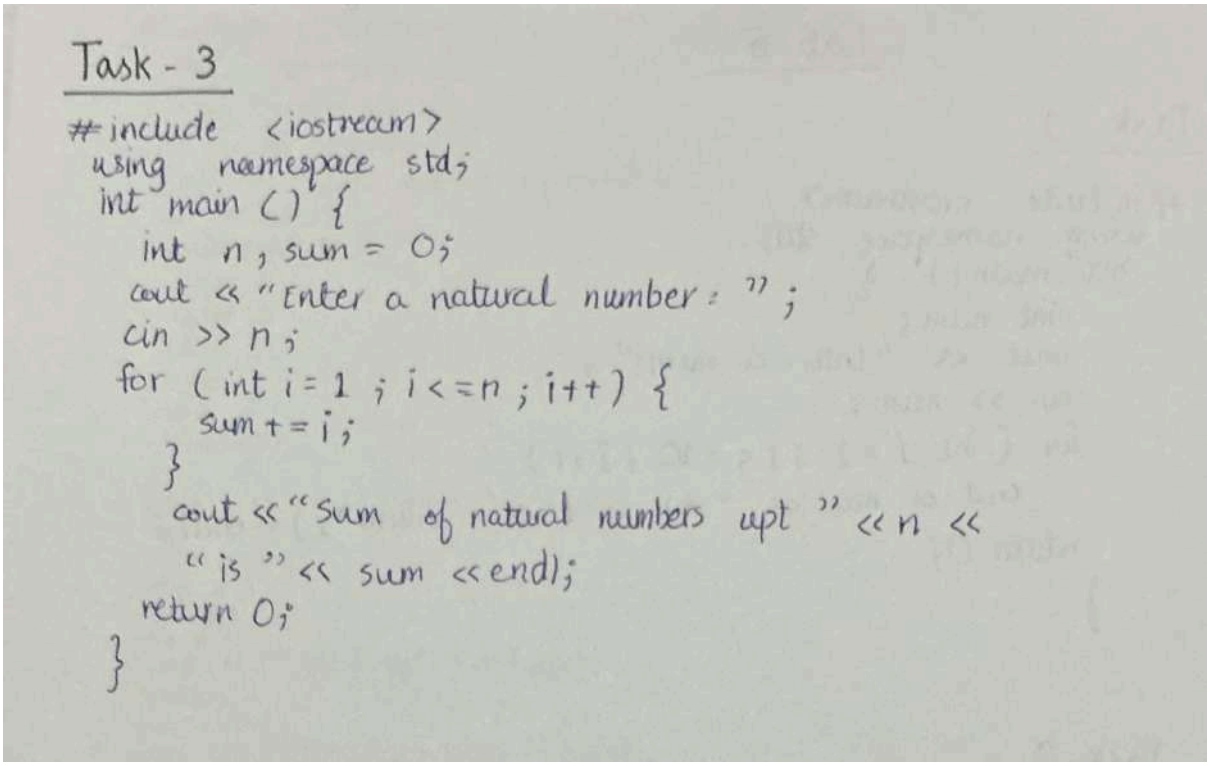
```
#include <iostream>
using namespace std;
int main() {
    int base, exp;
    int result = 1;
    cout << "Enter base: ";
    cin >> base;
    cout << "Enter exponent:" ;
    cin >> exp;
    for (int i=1; i <= exp; i++) {
      result = result * base;
    }
    cout << "Result: " << result << endl;
return 0;
}
```

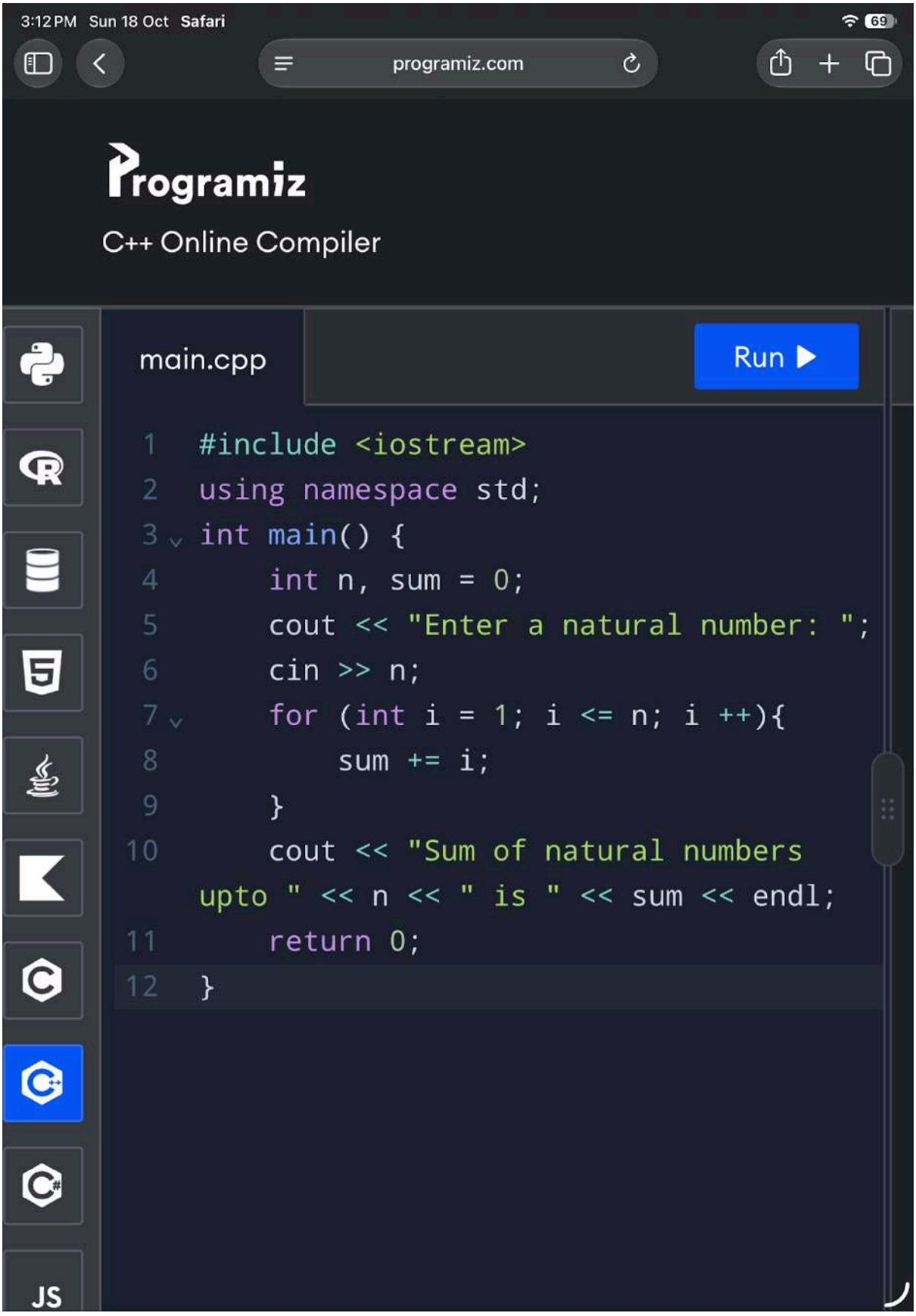

3:00 PM Sun 18 Oct Safari
programiz.com
C++ Online Compiler
main.cpp
Run
#include <iostream>
using namespace std;
int main() {
int base, exp;
int result = 1;
cout << "Enter base: ";
cin >> base;
cout << "Enter exponent: ";
cin >> exp;
for (int i = 1; i <= exp; i ++){
result = result * base;
}
cout << "Result: " << result << endl;
return 0;
}
JS

Output:

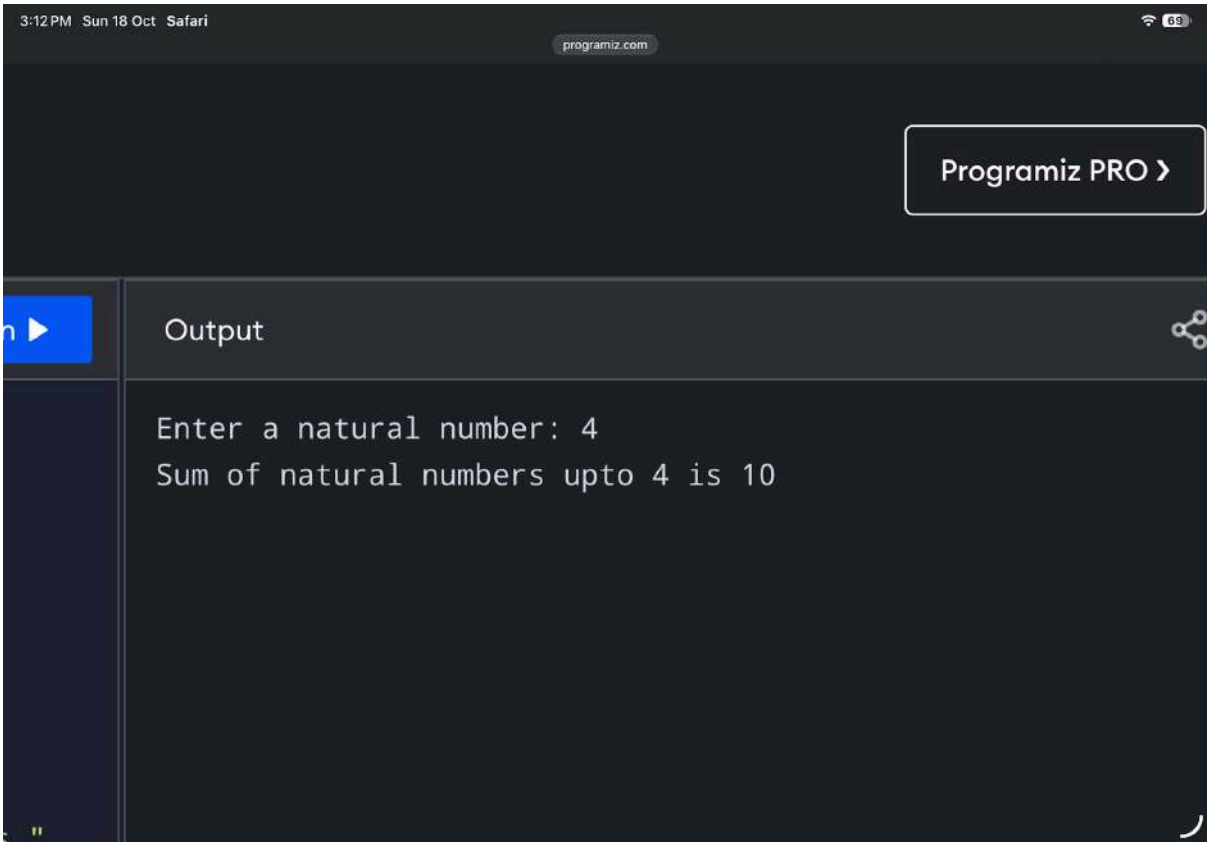

3:01 PM Sun 18 Oct Safari
programiz.com
Program
Run
Output
Enter base: 2
Enter exponent: 3
Result: 8

Task 3:


Task - 3
#include <iostream>
using namespace std;
int main () {
int n, sum = 0;
cout << "Enter a natural number: ";
cin >> n;
for (int i = 1 ; i <= n ; i++) {
sum += i;
}
cout << "Sum of natural numbers upt " << n <<
" is " << sum << endl;
return 0;
}


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Programiz
C++ Online Compiler
main.cpp
Run
#include <iostream>
using namespace std;
int main() {
int n, sum = 0;
cout << "Enter a natural number: ";
cin >> n;
for (int i = 1; i <= n; i ++){
sum += i;
}
cout << "Sum of natural numbers upto " << n << " is " << sum << endl;
return 0;
}
JS

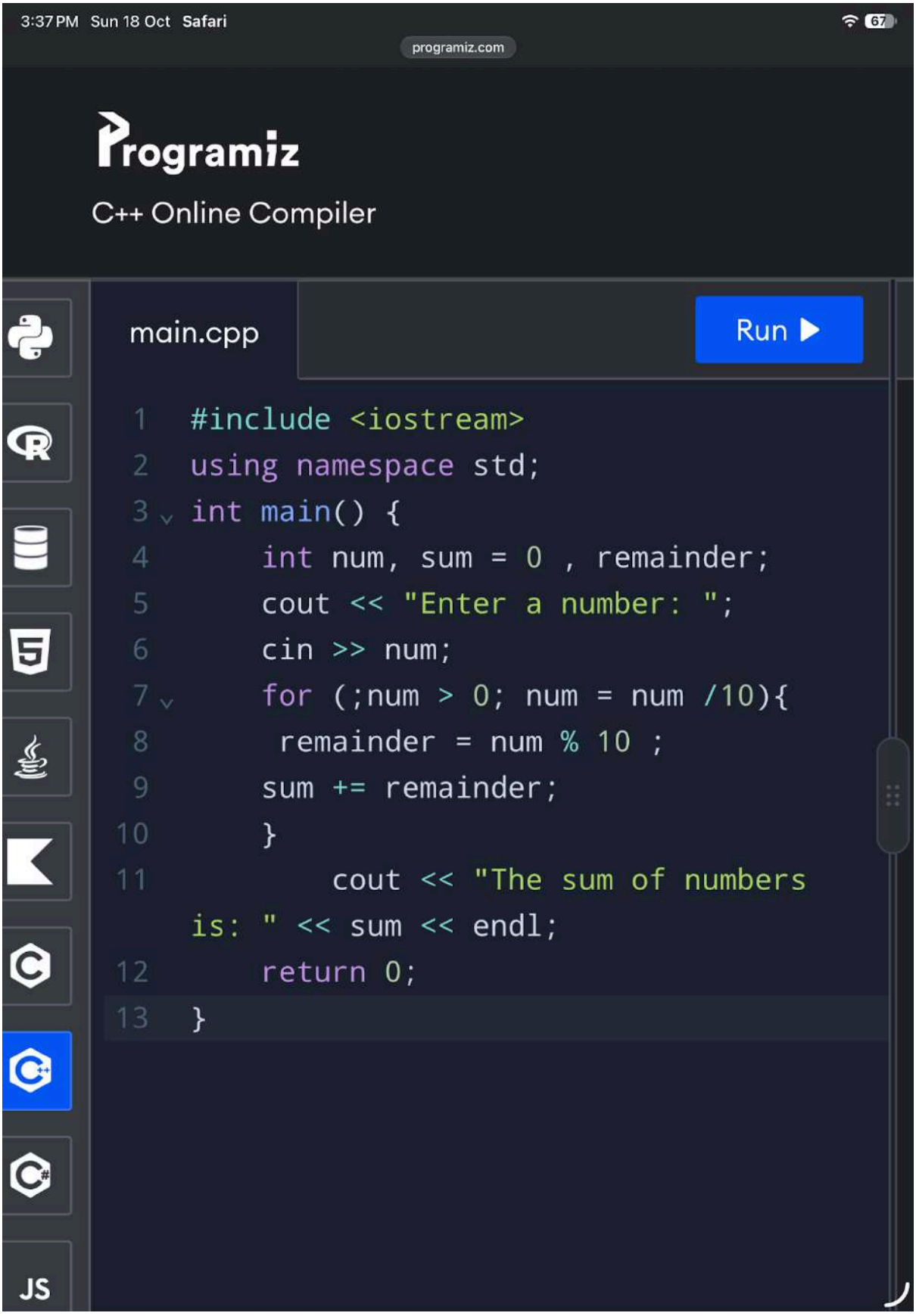
Output:



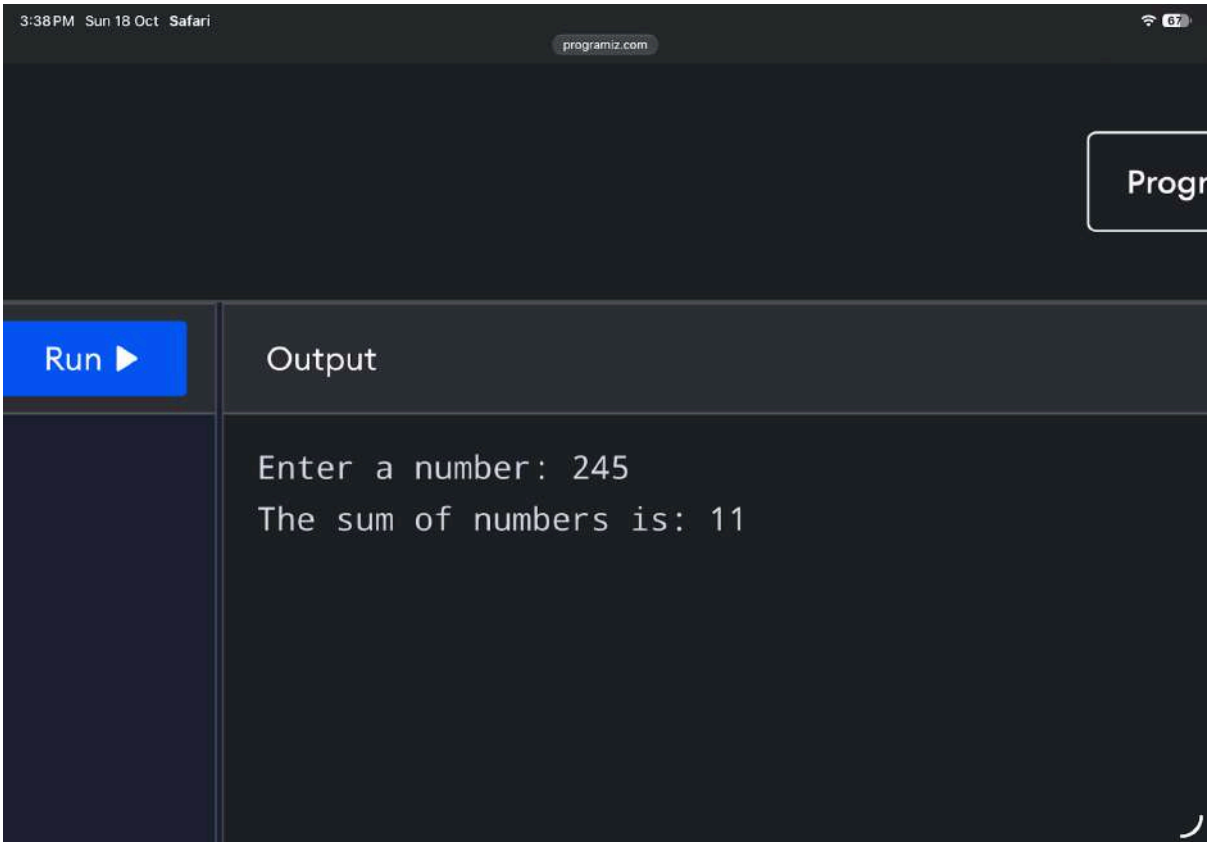
Task 4:

Task - 4

```
#include <iostream>
using namespace std;
int main() {
  int num, sum = 0, remainder;
  cout << "Enter a number: ";
  cin >> num;
  for (; num > 0; num = num/10) {
    remainder = num % 10;
    sum += remainder;
  }
  cout << "The sum of numbers is: " << sum << endl;
  return 0;
}
```


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67
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Programiz
C++ Online Compiler
main.cpp
Run
#include <iostream>
using namespace std;
int main() {
int num, sum = 0 , remainder;
cout << "Enter a number: ";
cin >> num;
for (;num > 0; num = num /10){
remainder = num % 10 ;
sum += remainder;
}
cout << "The sum of numbers is: " << sum << endl;
return 0;
}
JS

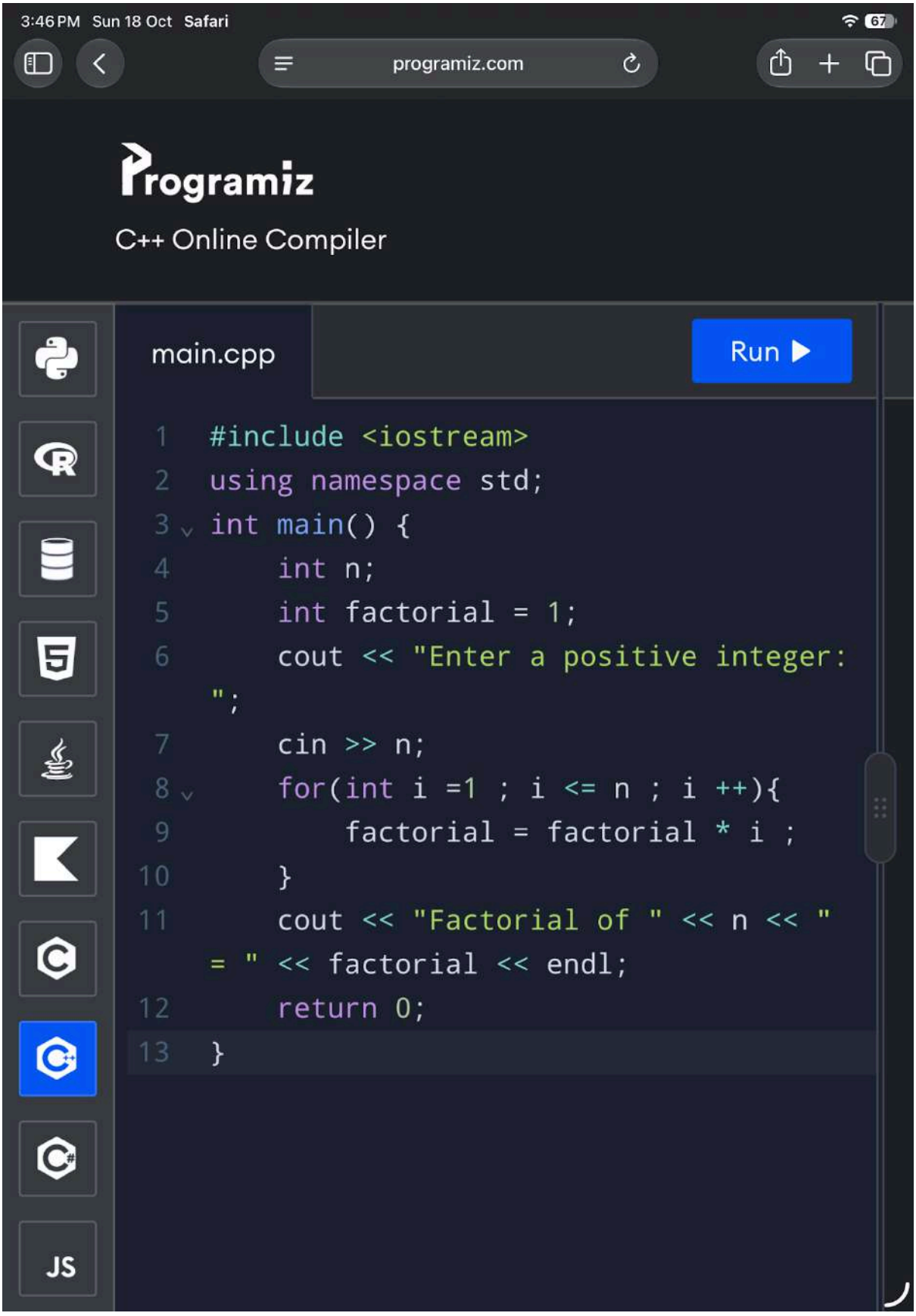
Output:



Task 5:

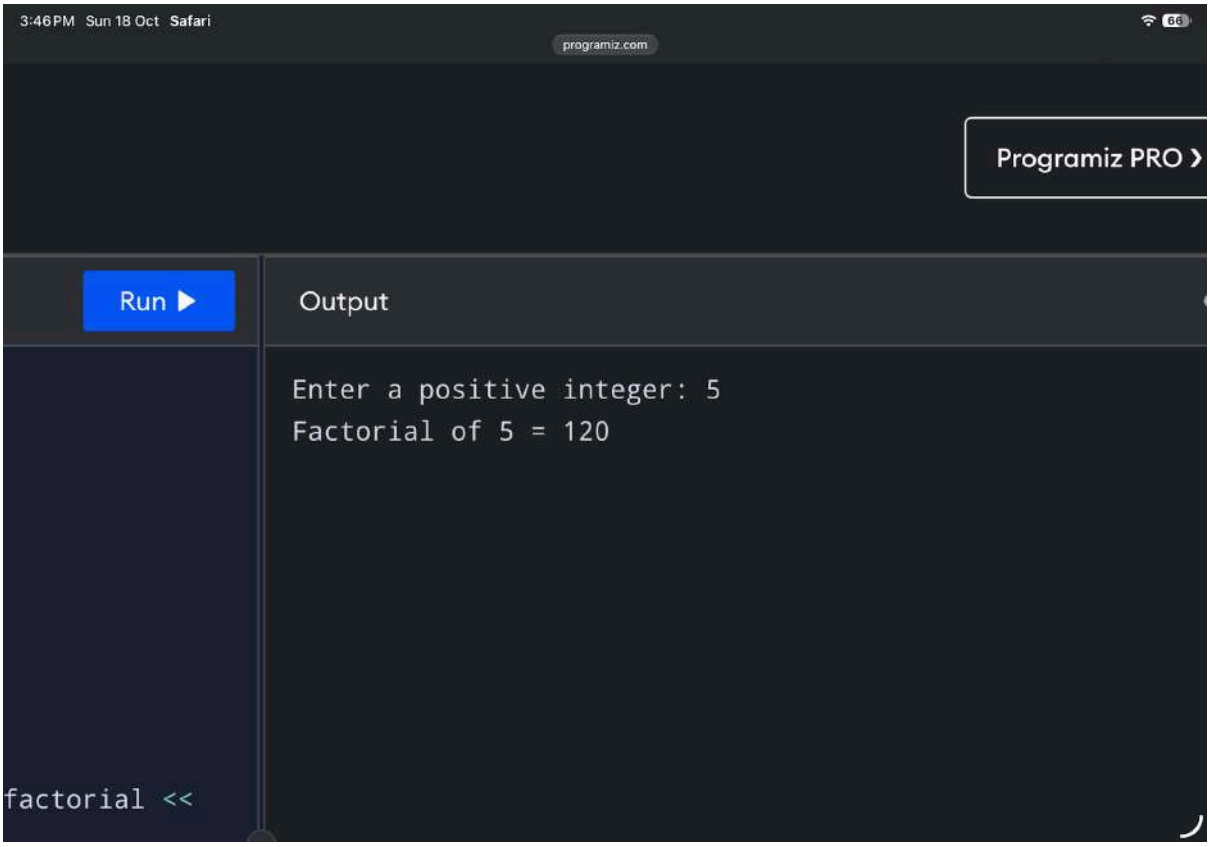
Task - 5

```
# include <iostream>
using namespace std;
int main () {
    int n;
    int factorial = 1;
    cout << " Enter a positive integer : ";
    cin >> n;
   for (int i = 1 ; i <= n ; i++) {
      factorial = factorial * i ;
      }
    cout << "Factorial of " << n << " = " << factorial << endl;
   return 0;
    }
```




```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    int factorial = 1;
    cout << "Enter a positive integer: ";
    cin >> n;
    for(int i =1 ; i <= n ; i ++){
        factorial = factorial * i ;
    }
    cout << "Factorial of " << n << " = " << factorial << endl;
    return 0;
}
```

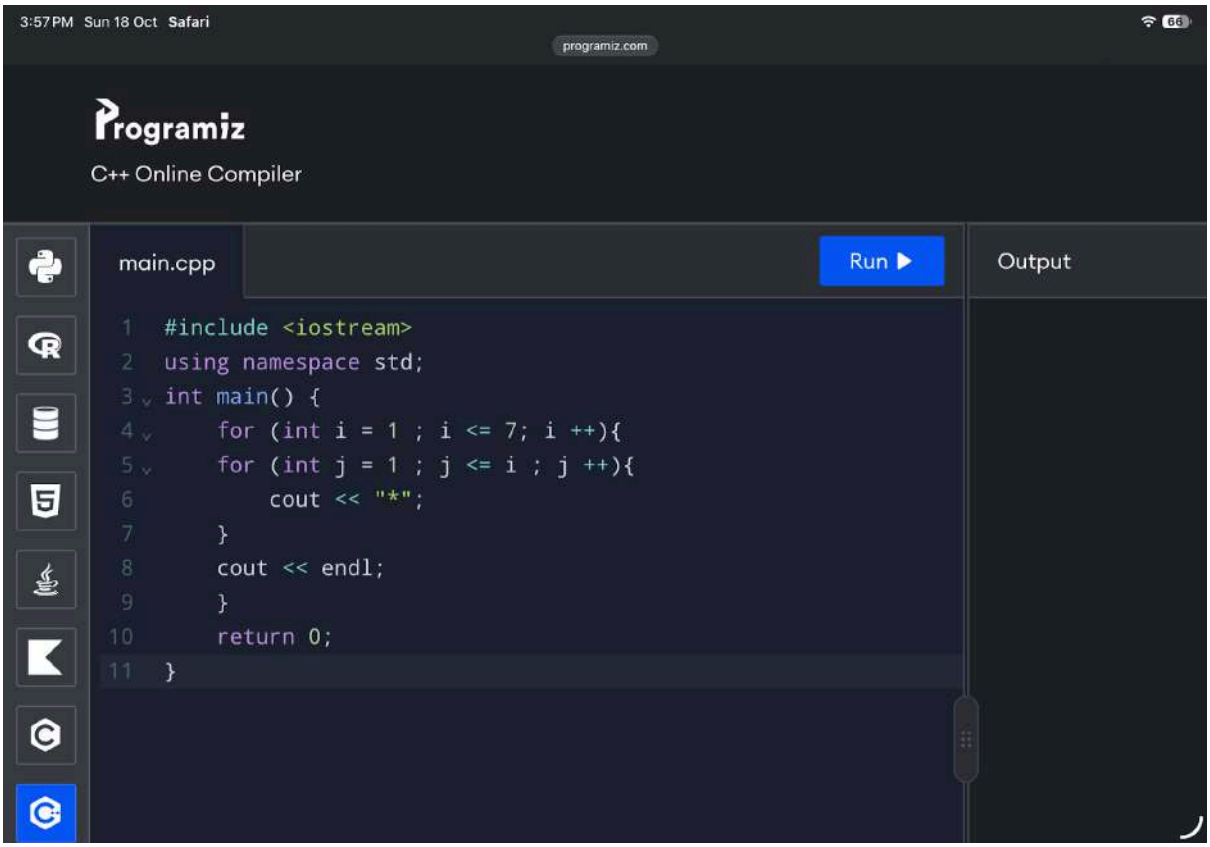
Output:


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Programiz PRO
Run
Output
Enter a positive integer: 5
Factorial of 5 = 120
factorial <<

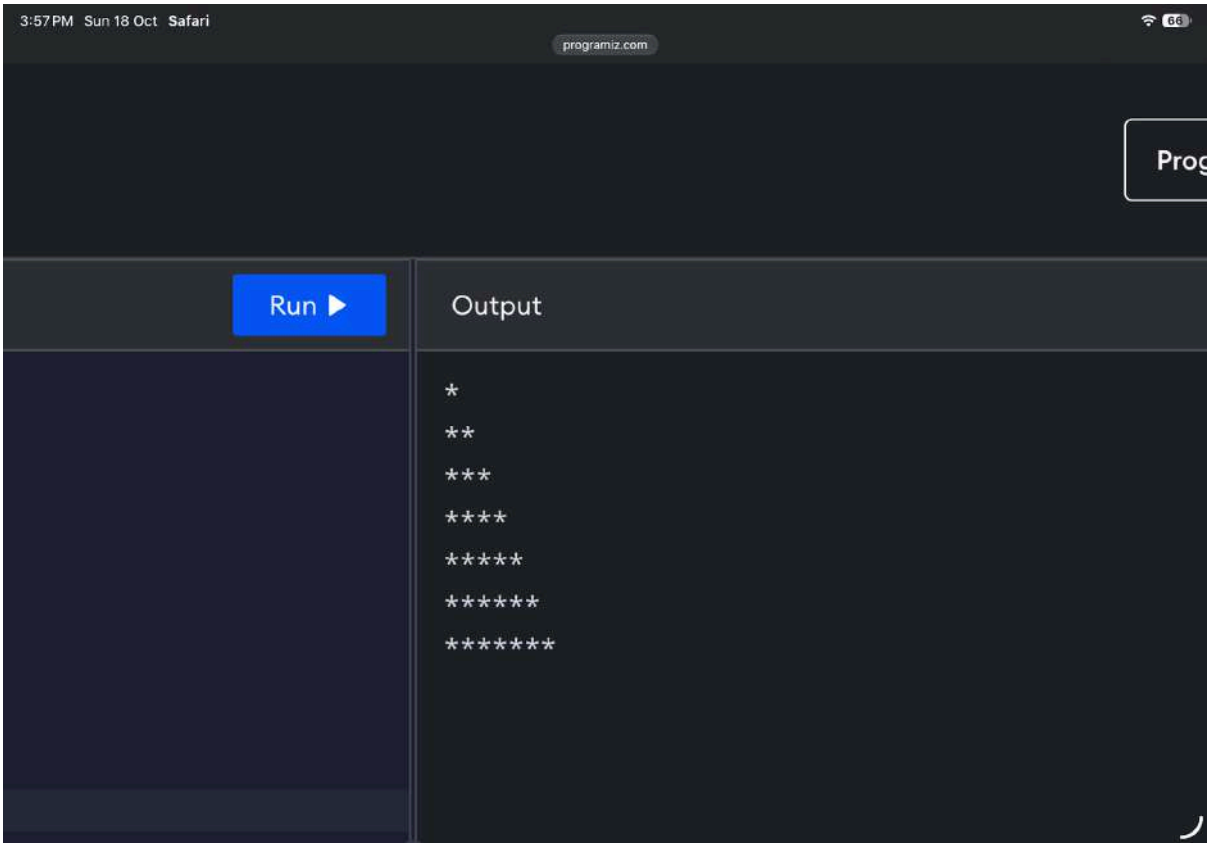
Task 7:

Task - 7

```
#include <iostream>
using namespace std;
int main () {
for (int i = 1 ; i <= 7 ; i++ ){
for (int j = 1 ; j <= i ; j++) {
cout << "*" ;
}
cout << endl;
}
return 0;
}
```



Output:



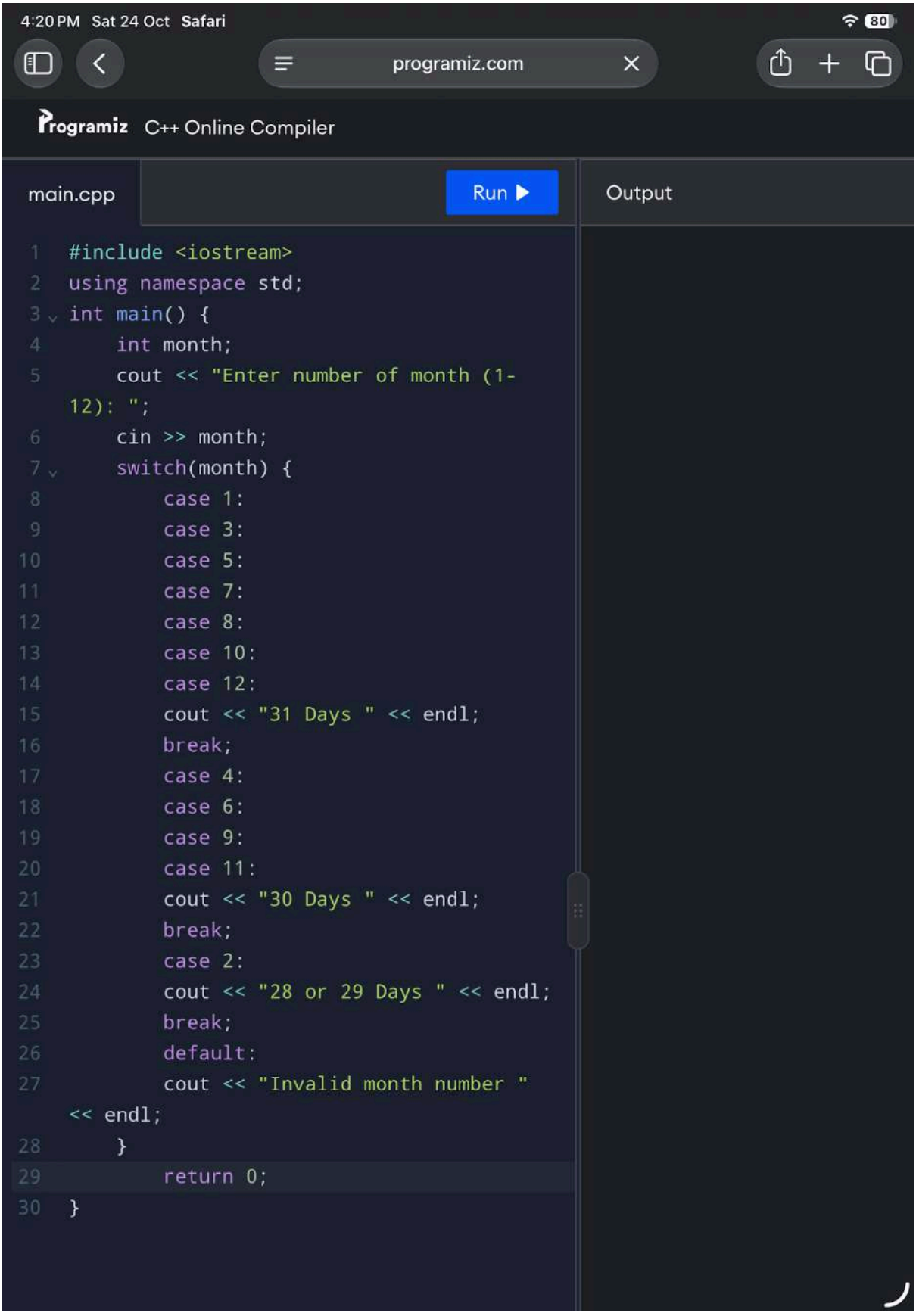
Switch Case Tasks

Task 1:

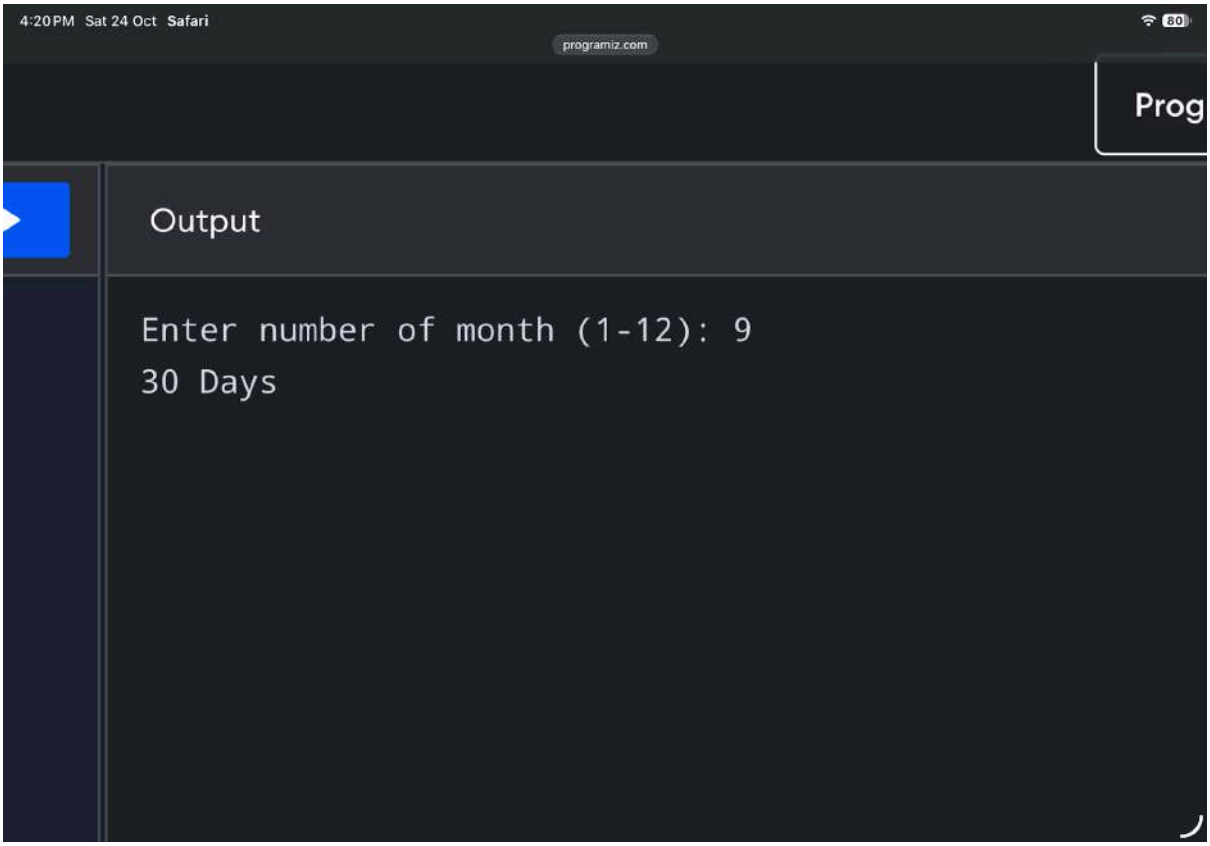
Switch Case Tasks

Number of Days in a month:

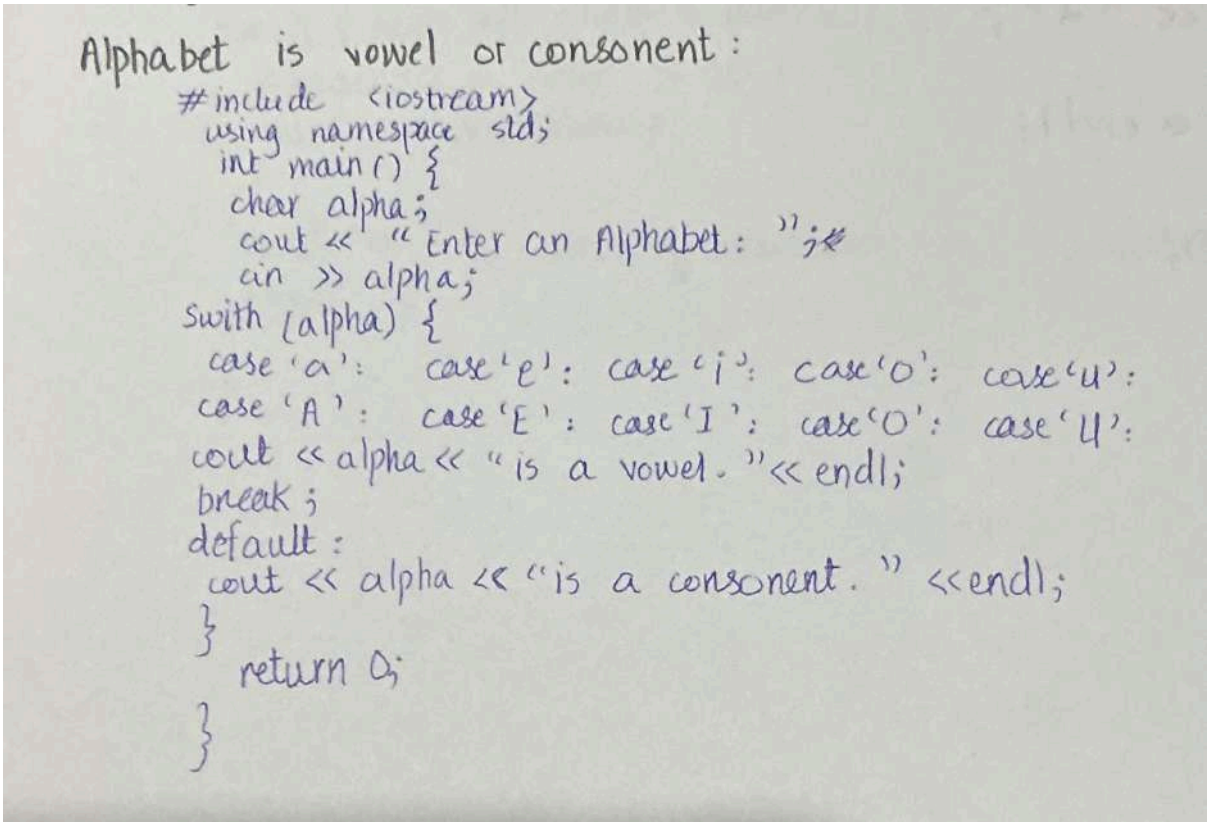
```
#include <iostream>
using namespace std;
int main () {
    int month;
cout << "Enter number of, month (1-12): ";
cin >> month;
swith (month) {
 case 1:
 casse 3:
 case 5:
 case 7:
 case 8:
 case 10:
 case 12:
 cout << "31 Days" << endl;
 break;
 case 4:
 case 6:
 case 9:
 case 11:
 cout << "30 Days" << endl;
 break;
 case 2:
 cout << "28 or 29 Days" << endl;
 break;
default:
cout << "Invalid month number" << endl;
}
 return 0;
}
```


4:20 PM Sat 24 Oct Safari
programiz.com
Programiz C++ Online Compiler
main.cpp
Run
Output
#include <iostream>
using namespace std;
int main() {
int month;
cout << "Enter number of month (1-12): ";
cin >> month;
switch(month) {
case 1:
case 3:
case 5:
case 7:
case 8:
case 10:
case 12:
cout << "31 Days " << endl;
break;
case 4:
case 6:
case 9:
case 11:
cout << "30 Days " << endl;
break;
case 2:
cout << "28 or 29 Days " << endl;
break;
default:
cout << "Invalid month number " << endl;
}
return 0;
}

Output:


4:20 PM Sat 24 Oct Safari
programiz.com
Prog
Output
Enter number of month (1-12): 9
30 Days

Task 2:


Alphabet is vowel or consonent:
#include <iostream>
using namespace std;
int main() {
char alpha;
cout << "Enter an Alphabet: ";
cin >> alpha;
swith (alpha) {
case 'a': case 'e': case 'i': case 'o': case 'u':
case 'A': case 'E': case 'I': case 'O': case 'U':
cout << alpha << "is a vowel." << endl;
break;
default:
cout << alpha << "is a consonent." << endl;
}
return 0;
}

```cpp
#include <iostream>
using namespace std;
int main() {
    char alpha;
    cout << "Enter an Alphabet: ";
    cin >> alpha;
    switch(alpha) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
        cout << alpha << " is a vowel." << endl;
        break;
        default:
        cout << alpha << " is a consonent." << endl;
    }
        return 0;
}
```

Output:

```
Enter an Alphabet: e
e is a vowel.
```

Think Tasks

What will be output of the following programs?

Task 1:

main.cpp

Run

```cpp
#include <iostream>
using namespace std;
int main() {
    for(5;2;2)
        {
            cout << "Hello";
        }
    return 0;
}
```

Output:

Run

Output

```
HelloHelloHelloH
```

main.cpp

Run ▶

Crop

Outpu

```cpp
#include <iostream>
using namespace std;
int main() {
    int i;
    for (i = 0; i>9; i+=3)
        {
            cout << "for";
        }
    return 0;
}
```

Output:

Run ▶

Output

Crop

Task 3:

main.cpp

Run

```cpp
#include <iostream>
using namespace std;
int main() {
    for ( ; ;)
        {
            cout << "10";
        }
    return 0;
}
```

Output:

Output

Task 4:

main.cpp Run ▶ Output

```cpp
#include <iostream>
using namespace std;
int main() {
    int fun = 5;
    cout << "C++ for loop ";
    int x = 5;
    for (x = 0; x <= fun; x++){
        cout << x;
    }
    return 0;
}
```

C++ for

Output:

Crop

Output

```
C++ for loop 012345
```
